Multi-Turn Speech Dialogue: Pre-2023 Supplement

A curated supplement of papers published in 2022 or earlier, excluding titles already listed in readme.docx.

Scope: multi-turn spoken interaction, spoken dialogue corpora and evaluation, turn-taking/interruption, incremental or barge-in-capable systems, and training methods for spoken dialogue. Text-only dialogue papers and single-turn ASR/TTS/SLU work are excluded unless they directly model turn-taking for spoken systems.

Cutoff	New papers	Deduplication
2022 and earlier	92	0 existing title(s) removed

Survey

1. **"Spoken Dialogue Technology: Enabling the Conversational User Interface".** *Michael F. McTear*. ACM Computing Surveys 2002. [Paper]
2. **"POMDP-Based Statistical Spoken Dialog Systems: A Review".** *Steve Young et al.*. Proceedings of the IEEE 2013. [Paper]
3. **"Using Neural Networks for Data-Driven Backchannel Prediction: A Survey on Input Features and Training Techniques".** *Markus Mueller et al.*. HCI International 2015. [Paper]
4. **"A Survey of Available Corpora For Building Data-Driven Dialogue Systems: The Journal Version".** *Iulian V. Serban et al.*. Dialogue & Discourse 2018. [Paper]
5. **"Turn-taking in Conversational Systems and Human-Robot Interaction: A Review".** *Gabriel Skantze*. Computer Speech & Language 2021. [Paper]

General Surveys

1. **"A Survey on Dialogue Systems: Recent Advances and New Frontiers".** *Hongshen Chen et al.*. SIGKDD Explorations 2017. [Paper]
2. **"Neural Approaches to Conversational AI".** *Jianfeng Gao et al.*. Foundations and Trends in Information Retrieval 2019. [Paper]
3. **"Recent Neural Methods on Dialogue State Tracking for Task-Oriented Dialogue Systems: A Survey".** *Vevake Balaraman et al.*. SIGDIAL 2021. [Paper]
4. **"Who Says What to Whom: A Survey of Multi-Party Conversations".** *Jia-Chen Gu et al.*. IJCAI 2022. [Paper]

Datasets

1. **"The HCRC Map Task Corpus".** *Anne H. Anderson et al.*. Language and Speech 1991. [Paper]
2. **"SWITCHBOARD: Telephone Speech Corpus for Research and Development".** *John J. Godfrey et al.*. ICASSP 1992. [Paper]
3. **"The ICSI Meeting Corpus".** *Adam Janin et al.*. ICASSP 2003. [Paper]
4. **"The Fisher Corpus: A Resource for the Next Generations of Speech-to-Text".** *Christopher Cieri et al.*. LREC 2004. [Paper]
5. **"The AMI Meeting Corpus".** *Wessel Kraaij et al.*. MLMI 2005. [Paper]
6. **"HKUST/MTS: A Very Large Scale Mandarin Telephone Speech Corpus".** *Yi Liu et al.*. ISCSLP 2006. [Paper]
7. **"IEMOCAP: Interactive Emotional Dyadic Motion Capture Database".** *Carlos Busso et al.*. Language Resources and Evaluation 2008. [Paper]

8. **"The SEMAINE Database: Annotated Multimodal Records of Emotionally Colored Conversations between a Person and a Limited Agent".** *Gary McKeown et al.*. IEEE Transactions on Affective Computing 2012. [Paper]
9. **"The MAHNOB Mimicry Database: A Database of Naturalistic Human Interactions".** *Sanjay Bilakhia et al.*. Pattern Recognition Letters 2015. [Paper]
10. **"The NoXi Database: Multimodal Recordings of Mediated Novice-Expert Interactions".** *Angelo Cafaro et al.*. ICMI 2017. [Paper]
11. **"Japanese Dialogue Corpus of Information Navigation and Attentive Listening Annotated with Extended ISO-24617-2 Dialogue Act Tags".** *Koichiro Yoshino et al.*. LREC 2018. [Paper]
12. **"Taskmaster-1: Toward a Realistic and Diverse Dialog Dataset".** *Bill Byrne et al.*. EMNLP-IJCNLP 2019. [Paper]
13. **"Alexa in the Wild: Collecting Unconstrained Conversations with a Modern Voice Assistant in a Public Environment".** *Benjamin Benk et al.*. LREC 2020. [Paper]
14. **"HarperValleyBank: A Domain-Specific Spoken Dialog Corpus".** *Mike Wu et al.*. arXiv 2020. [Paper]
15. **"The MSP-Conversation Corpus".** *Luz Martinez-Lucas et al.*. Interspeech 2020. [Paper]
16. **"ASCEND: A Spontaneous Chinese-English Dataset for Code-switching in Multi-turn Conversation".** *Holy Lovenia et al.*. arXiv 2021. [Paper]
17. **"Collection and Analysis of Travel Agency Task Dialogues with Age-Diverse Speakers".** *Michimasa Inaba et al.*. LREC 2022. [Paper]
18. **"Design and Evaluation of the Corpus of Everyday Japanese Conversation".** *Hanae Koiso et al.*. LREC 2022. [Paper]

Evaluation & Benchmarks

Evaluation Focus: Turn-taking & Interruption

1. **"Effects of System Barge-In Responses on User Impressions".** *Jun-ichi Hirasawa et al.*. Eurospeech 1999. [Paper]
2. **"Learning Decision Trees to Determine Turn-Taking by Spoken Dialogue Systems".** *Ryo Sato et al.*. ICSLP 2002. [Paper]
3. **"A Finite-State Turn-Taking Model for Spoken Dialog Systems".** *Antoine Raux and Maxine Eskenazi*. NAACL-HLT 2009. [Paper]
4. **"Multimodal End-of-Turn Prediction in Multi-Party Meetings".** *Iwan de Kok and Dirk Heylen*. ICMI-MLMI 2009. [Paper]
5. **"Evaluation and Optimisation of Incremental Processors".** *Okko Buss and David Schlangen*. Dialogue & Discourse 2011. [Paper]
6. **"Turn-Taking Cues in Task-Oriented Dialogue".** *Agustin Gravano and Julia Hirschberg*. Computer Speech & Language 2011. [Paper]
7. **"Continuously Predicting and Processing Barge-in During a Live Spoken Dialogue Task".** *Ethan O. Selfridge et al.*. SIGDIAL 2013. [Paper]
8. **"Predicting User Satisfaction from Turn-Taking in Spoken Conversations".** *Shammur Absar Chowdhury et al.*. Interspeech 2016. [Paper]
9. **"Towards a General, Continuous Model of Turn-Taking in Spoken Dialogue Using LSTM Recurrent Neural Networks".** *Gabriel Skantze*. SIGDIAL 2017. [Paper]
10. **"Towards Deep End-of-Turn Prediction for Situated Spoken Dialogue Systems".** *Angelika Maier et al.*. Interspeech 2017. [Paper]

11. **"Turn-Taking Estimation Model Based on Joint Embedding of Lexical and Prosodic Contents".** *Chaoran Liu et al.*. Interspeech 2017. [Paper]
12. **"Improving End-of-Turn Detection in Spoken Dialogues by Detecting Speaker Intentions as a Secondary Task".** *Zakaria Aldeneh et al.*. ICASSP 2018. [Paper]
13. **"Investigating Speech Features for Continuous Turn-Taking Prediction Using LSTMs".** *Matthew Roddy et al.*. Interspeech 2018. [Paper]
14. **"Neural Dialogue Context Online End-of-Turn Detection".** *Ryo Masumura et al.*. SIGDIAL 2018. [Paper]
15. **"Prediction of Turn-Taking Using Multitask Learning with Prediction of Backchannels and Fillers".** *Kohei Hara et al.*. Interspeech 2018. [Paper]
16. **"Turn-Taking Predictions Across Languages and Genres Using an LSTM Recurrent Neural Network".** *Nigel G. Ward et al.*. SLT 2018. [Paper]
17. **"An Incremental Turn-Taking Model for Task-Oriented Dialog Systems".** *Andrei C. Coman et al.*. Interspeech 2019. [Paper]
18. **"Turn-Taking Prediction Based on Detection of Transition Relevance Place".** *Kohei Hara et al.*. Interspeech 2019. [Paper]
19. **"Oh, Jeez! or Uh-Huh? A Listener-Aware Backchannel Predictor on ASR Transcriptions".** *Daniel Ortega et al.*. ICASSP 2020. [Paper]
20. **"TurnGPT: A Transformer-Based Language Model for Predicting Turn-Taking in Spoken Dialog".** *Erik Ekstedt and Gabriel Skantze.* EMNLP Findings 2020. [Paper]
21. **"BPM_MT: Enhanced Backchannel Prediction Model Using Multi-Task Learning".** *Jin Yea Jang et al.*. EMNLP 2021. [Paper]
22. **"Projection of Turn Completion in Incremental Spoken Dialogue Systems".** *Erik Ekstedt and Gabriel Skantze.* SIGDIAL 2021. [Paper]
23. **"Timing Generating Networks: Neural Network Based Precise Turn-Taking Timing Prediction in Multiparty Conversation".** *Shinya Fujie et al.*. Interspeech 2021. [Paper]
24. **"Contextual Acoustic Barge-In Classification for Spoken Dialog Systems".** *Dhanush Bekal et al.*. Interspeech 2022. [Paper]
25. **"Gated Multimodal Fusion with Contrastive Learning for Turn-Taking Prediction in Human-Robot Dialogue".** *Jiudong Yang et al.*. ICASSP 2022. [Paper]
26. **"Response Timing Estimation for Spoken Dialog System Using Dialog Act Estimation".** *Jin Sakuma et al.*. Interspeech 2022. [Paper]
27. **"Turn-Taking Prediction for Natural Conversational Speech".** *Shuo-Yiin Chang et al.*. Interspeech 2022. [Paper]
28. **"Voice Activity Projection: Self-Supervised Learning of Turn-Taking Events".** *Erik Ekstedt and Gabriel Skantze.* Interspeech 2022. [Paper]

Evaluation Focus: Full-duplex Interaction

No additional eligible pre-2023 paper was found after deduplication. Early work in this area is represented under Half-duplex / Controlled Barge-in rather than modern native full-duplex interaction.

Evaluation Focus: Multi-turn Dialogue Capabilities

1. **"PARADISE: A Framework for Evaluating Spoken Dialogue Agents".** *Marilyn A. Walker et al.*. ACL-EACL 1997. [Paper]
2. **"Quantitative and Qualitative Evaluation of DARPA Communicator Spoken Dialogue Systems".** *Julie E. Boland et al.*. ACL 2001. [Paper]

3. **"The PARADISE Evaluation Framework: Issues and Findings".** *Melita Hajdinjak and France Mihelic*. Computational Linguistics 2006. [Paper]
4. **"A Framework for Model-Based Evaluation of Spoken Dialog Systems".** *Sebastian Moller et al.*. SIGDIAL 2008. [Paper]
5. **"User Simulation as Testing for Spoken Dialog Systems".** *Hua Ai and Fuliang Weng*. SIGDIAL 2008. [Paper]
6. **"Spoken Dialog Challenge 2010: Comparison of Live and Control Test Results".** *Alan W. Black et al.*. SIGDIAL 2011. [Paper]
7. **"Position Paper: Towards Standardized Metrics and Tools for Spoken and Multimodal Dialog System Evaluation".** *Sebastian Moller et al.*. NAACL Workshop 2012. [Paper]
8. **"The Dialog State Tracking Challenge".** *Jason D. Williams et al.*. SIGDIAL 2013. [Paper]
9. **"The Second Dialog State Tracking Challenge".** *Matthew Henderson et al.*. SIGDIAL 2014. [Paper]

Models

Interaction Mode: Turn-taking

1. **"A Robust System for Natural Spoken Dialogue".** *James F. Allen et al.*. ACL 1996. [Paper]
2. **"Galaxy-II: A Reference Architecture for Conversational System Development".** *Stephanie Seneff et al.*. ICSLP 1998. [Paper]
3. **"Let's Go Public! Taking a Spoken Dialog System to the Real World".** *Antoine Raux et al.*. Interspeech 2005. [Paper]
4. **"The RavenClaw Dialog Management Framework: Architecture and Systems".** *Dan Bohus and Alexander I. Rudnicky*. Computer Speech & Language 2009. [Paper]
5. **"The SEMAINE API: Towards a Standards-Based Framework for Building Emotion-Oriented Systems".** *Marc Schroder*. Advances in Human-Computer Interaction 2010. [Paper]
6. **"Attentive Listening System with Backchanneling, Response Generation and Flexible Turn-Taking".** *Divesh Lala et al.*. SIGDIAL 2017. [Paper]
7. **"Towards End-to-End Spoken Dialogue Systems with Turn Embeddings".** *Ali Orkan Bayer et al.*. Interspeech 2017. [Paper]

Interaction Mode: Half-duplex / Controlled Barge-in

1. **"A Multiparty Multimodal Architecture for Realtime Turntaking".** *Kristinn R. Thorisson et al.*. IVA 2010. [Paper]
2. **"Towards Incremental Speech Generation in Dialogue Systems".** *Gabriel Skantze and Anna Hjalmarsson*. SIGDIAL 2010. [Paper]
3. **"A General, Abstract Model of Incremental Dialogue Processing".** *David Schlangen and Gabriel Skantze*. Dialogue & Discourse 2011. [Paper]
4. **"A Demonstration of Incremental Speech Understanding and Confidence Estimation in a Virtual Human Dialogue System".** *David DeVault and David Traum*. SIGDIAL 2012. [Paper]
5. **"The InproTK 2012 Release".** *Timo Baumann and David Schlangen*. NAACL Workshop 2012. [Paper]
6. **"An Easy Method to Make Dialogue Systems Incremental".** *Hatim Khouzaimi et al.*. SIGDIAL 2014. [Paper]
7. **"Incremental Dialog Processing in a Task-Oriented Dialog".** *Fabrizio Ghigi et al.*. Interspeech 2014. [Paper]

Interaction Mode: Full-duplex

No additional eligible pre-2023 paper was found after deduplication. Early work in this area is represented under Half-duplex / Controlled Barge-in rather than modern native full-duplex interaction.

Training Methods

Interaction Mode: Turn-taking

1. **"NJFun: A Reinforcement Learning Spoken Dialogue System".** *Diane Litman et al.*. ANLP-NAACL Workshop 2000. [Paper]
2. **"Optimizing Dialogue Management with Reinforcement Learning: Experiments with the NJFun System".** *Satinder Singh et al.*. Journal of Artificial Intelligence Research 2002. [Paper]
3. **"Agenda-Based User Simulation for Bootstrapping a POMDP Dialogue System".** *Jost Schatzmann et al.*. HLT-NAACL 2007. [Paper]
4. **"Partially Observable Markov Decision Processes for Spoken Dialog Systems".** *Jason D. Williams and Steve Young*. Computer Speech & Language 2007. [Paper]
5. **"Learning from Real Users: Rating Dialogue Success with Neural Networks for Reinforcement Learning in Spoken Dialogue Systems".** *Pei-Hao Su et al.*. Interspeech 2015. [Paper]
6. **"A Sequence-to-Sequence Model for User Simulation in Spoken Dialogue Systems".** *Layla El Asri et al.*. Interspeech 2016. [Paper]
7. **"End-to-End LSTM-Based Dialog Control Optimized with Supervised and Reinforcement Learning".** *Jason D. Williams and Geoffrey Zweig*. arXiv 2016. [Paper]
8. **"On-Line Active Reward Learning for Policy Optimisation in Spoken Dialogue Systems".** *Pei-Hao Su et al.*. ACL 2016. [Paper]
9. **"Deep Reinforcement Learning of Dialogue Policies with Less Weight Updates".** *Heriberto Cuayahuitl and Seunghak Yu*. Interspeech 2017. [Paper]
10. **"Towards Generalized Models for Task-Oriented Dialogue Modeling on Spoken Conversations".** *Ruijie Yan et al.*. arXiv 2022. [Paper]

Interaction Mode: Half-duplex / Streaming

1. **"Inverse Reinforcement Learning for Micro-Turn Management".** *Dongho Kim et al.*. Interspeech 2014. [Paper]
2. **"An Incremental Turn-Taking Model with Active System Barge-In for Spoken Dialog Systems".** *Tiancheng Zhao et al.*. SIGDIAL 2015. [Paper]
3. **"Reinforcement Learning for Turn-Taking Management in Incremental Spoken Dialogue Systems".** *Hatim Khouzaimi et al.*. IJCAI 2016. [Paper]
4. **"A Methodology for Turn-Taking Capabilities Enhancement in Spoken Dialogue Systems Using Reinforcement Learning".** *Hatim Khouzaimi et al.*. Computer Speech & Language 2018. [Paper]

Interaction Mode: Full-duplex

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